

Where does attention go, and when?

Phase-resolved spatial routing in a noisy-memory Luo–Maunsell change-detection agent

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RViT+ attention evidence program · 4 August 2026

Abstract

A recurrent vision transformer with a noisy spatial recurrent memory (mnemonic noise SD 0.64) was trained by reinforcement learning on a seven-frame analogue of the Luo and Maunsell (2015) two-location orientation change-detection task, which contains no spatial cue. We re-express the frozen checkpoint’s cross-attention weights in the common-quadrant convention used throughout our value-directed-attention (VDA) series and resolve them jointly by *frame* and by *environment condition* (changed vs. unchanged first test $\times$ tested location L0 vs. L3), holding the latent trial fixed and averaging only over 64 repetitions of sensory and mnemonic noise. Three results follow. (i) The two key sources dissociate completely: the visual-key map is *identical in all four conditions at every frame* and is governed only by which patches contain a Gabor, which it consistently *avoids* (0.215 vs. 0.285 conditional quadrant mass during the samples; 0.201 vs. 0.267 at the tests). (ii) All time-varying, condition-contingent selectivity lives in the recurrent-memory keys, which follow a four-stage encode $\rightarrow$ maintain $\rightarrow$ reset $\rightarrow$ select sequence: uniform at t_0 , split across both sample quadrants at t_1 – t_2 , returned to near-uniform at first-test onset t_3 , and concentrated on the tested location at t_4 (0.49–0.57 vs. 0.14–0.17) and through the gap t_5 (0.84–0.91). (iii) That selection is not location-general: at the guaranteed second test t_6 memory routing collapses onto a *fixed* quadrant (L0, 0.70 in both conditions), so the tested-minus-other contrast reverses sign between the two tested locations (+0.194 vs. –0.188). Memory keys carry only 1–14% of the joint softmax, and that share tracks the number of visible Gabors (0.010 blank, 0.075 one, 0.140 two) rather than the task phase. A companion suppression experiment drove sample-phase mass at the future-tested quadrant from 0.234 to 0.001 with zero discordant trials in 704 matched pairs, so none of this routing is shown to be response-causal. **Validity:** the checkpoint predates a corrected orientation-sampling contract and is retained for forensic provenance only (Section 5). This is a measurement and display report, not an admissible behavioural or biological claim.

1 Introduction

Luo and Maunsell (2015) dissociated attention-driven changes in *sensitivity* from changes in *criterion* by counterphasing the reward schedule across two stimulus locations in an orientation change-detection task that contains no visual cue. The task is attractive as a model system precisely because selection must be endogenous: both sample Gabors are simultaneously task-relevant, and nothing in the display tells the observer which location will be tested.

Our VDA series has so far studied *cued* change detection, where a spatial cue supplies a named location to which routing weights can be aligned. This report asks the complementary question in the uncued Luo task: in a trained agent, where does attention go at each frame, and does that allocation differ across the environment conditions the agent actually experiences? We measure the same quantities and use the same display convention as the VDA figures — a common physical 2×2 partition of the native key grid, key sources split before any query reduction, and uniform-normalized quadrant scores — so that the uncued and cued results are directly comparable.

The contribution is a measurement, not a mechanism. We show that in this checkpoint the visual and recurrent-memory key blocks play completely different roles, that the memory block alone carries the phase-resolved spatial signal, and that its apparent “return to the tested site” fails a location-generalizability control. We also report the negative causal result that accompanies it.

2 Methods

2.1 Task

The environment (luo2015_criterion) is a scaled seven-frame analogue of the published protocol. Two Gabors occupy diametrically opposed locations of a logical 2×2 grid, top-left (L0) and bottom-right (L3). Frames t_0 – t_1 present both samples; t_2 is a blank delay; t_3 – t_4 are the first-test response window at one randomly chosen location; t_5 is an inter-test gap; and t_6 is a guaranteed changed second test, presented only on trials whose first test was unchanged (Figure 1). Declaring during t_3 – t_4 scores a hit or a false alarm; withholding is verified by requiring a declaration to the second test. There is no cue at any point. Sensory orientation noise is 5.0° per drawn Gabor.

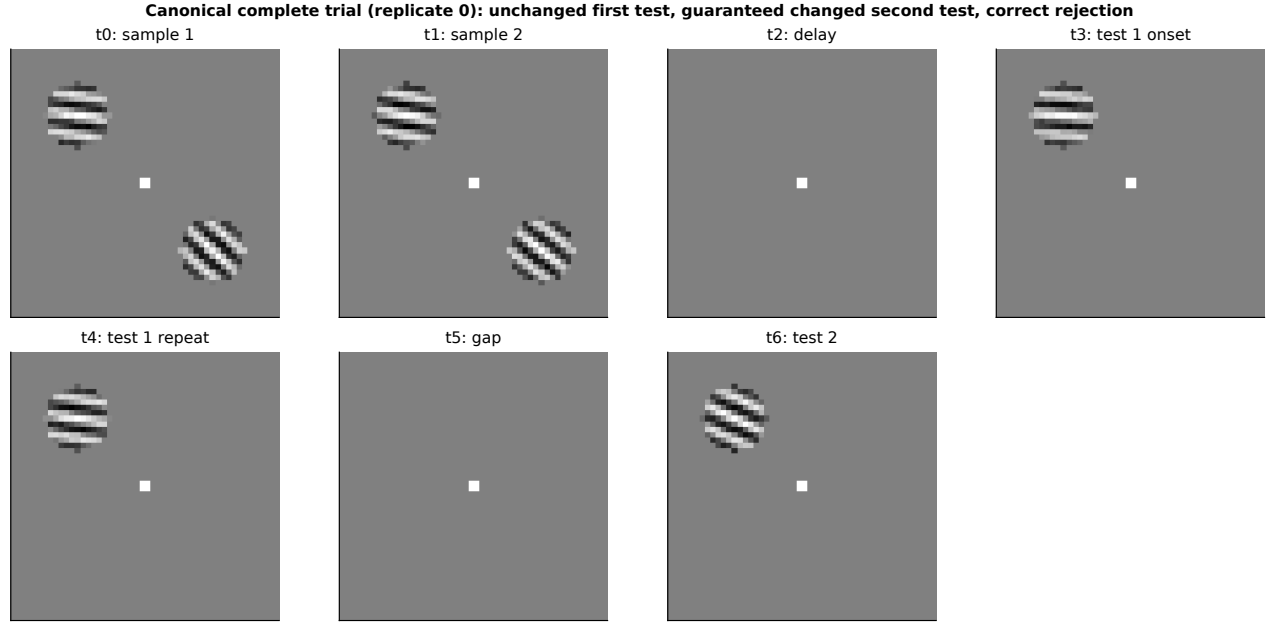


Figure 1: The seven-frame uncued task. A complete unchanged-first-test trial ending in a correct rejection at the guaranteed changed second test. Both samples appear at t_0 – t_1 ; the delay t_2 and gap t_5 are blank; one location is tested at t_3 – t_4 and again at t_6 . No cue appears at any frame.

2.2 Model and checkpoint

The agent is a recurrent vision transformer with a convolutional front end, a 20×20 patch grid over a 50×50 scene, an xLSTM spatial recurrent memory of width $d_{\text{mem}} = 32$ with decay 1.0 and additive mnemonic noise of SD 0.64, and an actor–critic head with two actions (wait, declare). Routing is `crossattn1`: current visual tokens supply the queries while visual tokens and recurrent-memory tokens jointly supply keys and values in a single softmax, so the key axis has $2 \times 400 = 800$ entries split into a 400-token visual block and a 400-token memory block. All results come from one frozen mid-training checkpoint (iteration 12,899, SHA-256 2c4a6d2d. . .), copied and hashed before analysis. Nothing was retrained or re-run for this report; the figures re-express arrays cached by the 2026-08-03 scientific assay.

2.3 Attention measurement and display convention

Let A be one trial’s 400×800 attention matrix at a named frame. Following the source-separated convention already used for VDA4, VDA16 and the spatial-scaling battery, we split the visual and memory key blocks *before* averaging over queries and before any spatial reduction. Averaging over queries gives per-source incoming token mass; summing that over each of the four quadrants of the 20×20 grid gives a source share w_s (the fraction of the joint softmax landing on source s , with $w_V + w_M = 1$) and, dividing by w_s , a *source-conditional quadrant total* that sums to one within each source. The uniform baseline for that score is therefore 0.25 per quadrant regardless of source, which is what makes the panels in Figures 3–4 comparable across frames, conditions and sources. We separately report query-resolved routing: the mass sent from the tested-location query quadrant to the same versus the other sample quadrant, within each source.

2.4 Conditions and trial construction

The design crosses first-test change status (changed, unchanged) with tested location (L0, L3), giving four environment conditions. Within each condition the latent trial is held fixed — same sample orientations, same tested location, same $+18^\circ$ signed change — and the only variation across the 64 trials per cell is the sensory and mnemonic noise. Policy actions are sampled, not argmaxed. Changed trials terminate inside the first-test window (declaration, or a miss at the end of t_4), so frames t_5 – t_6 are never experienced on those trials and are shown greyed out rather than averaged in. This is the survivor control that a naive frame-averaged map silently violates.

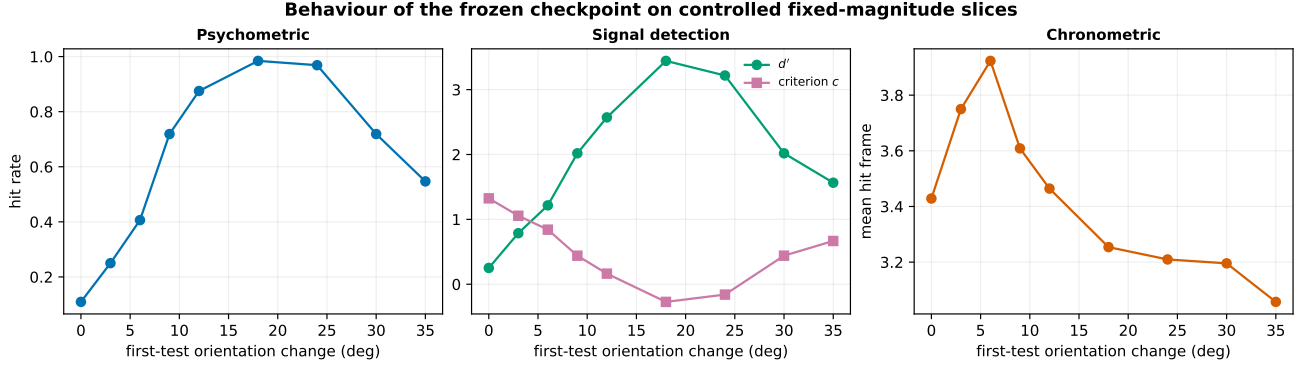


Figure 2: Behaviour on controlled fixed-magnitude slices. Hit rate, signal-detection decomposition, and mean hit frame as a function of first-test orientation change. Sensitivity peaks at 18° — the probe used for every attention figure — and both d' and c degrade at the largest changes.

3 Results

3.1 Behaviour of the frozen checkpoint

On the fixed +18° probe used for all attention figures the checkpoint is near ceiling: 63/64 and 64/64 hits at L0 and L3, and 60/64 and 59/64 correct rejections, i.e. 246/256 (0.961) correct. Across the full controlled magnitude ladder natural accuracy is 476/704 (0.676). The psychometric function is non-monotonic (Figure 2): hit rate and d' peak at 18° (0.984, $d' = 3.44$) and fall to 0.547 and $d' = 1.57$ at 35°, while the criterion c falls to -0.30 at 18° and rises again to +0.66 at 35°. Because sensitivity and criterion move together at the large-change end, the decline is not attributable to sensitivity alone, and we do not offer a mechanism for it here. *These are controlled fixed-magnitude slices within the training support, not the training distribution; each point is 64 changed trials against 128 shared no-change trials from one checkpoint.*

3.2 Key-source share tracks visible stimuli, not task phase

Memory keys receive a small and highly structured minority of the joint softmax. Their share w_M is 0.140 at t_0 – t_1 , 0.010 at t_2 and t_5 , and 0.075 at t_3 , t_4 and t_6 (middle row of Figure 5). These three values correspond exactly to two visible Gabors, none, and one. The share is identical in all four conditions to three decimals. **The visual/memory balance is therefore a function of how much stimulus energy is on the screen, not of what stage of the trial the agent is in** — the delay and the gap are indistinguishable to it, as are the first and second tests. Any statement about memory-key selectivity below is conditional on a source that never exceeds 14% of the total mass.

3.3 Visual keys: one fixed map that avoids the stimuli

The source-conditional visual map (Figure 3) is remarkable for what it does *not* do. Its four quadrant values are the same in all four conditions at every frame, to two decimals, and they depend only on which patches currently contain a Gabor:

- Both samples present (t_0 , t_1): stimulus quadrants 0.215, blank quadrants 0.280 and 0.290.
- Blank frames (t_2 , t_5): 0.250 everywhere, i.e. exactly uniform.
- One Gabor present (t_3 , t_4 , t_6): tested quadrant 0.201, the other three 0.267.

The sign is consistent and against the naive expectation: incoming visual-key attention is *lower* at occupied locations than at empty ones, by 0.065–0.066 of conditional quadrant mass. The query-resolved version is equally rigid — the tested-location query sends 0.048 *less* visual mass to its own quadrant than to the other one, at every test frame in every condition (bottom row of Figure 5). This reproduces, in the common-quadrant display, the negative sample-stimulus-minus-blank (−0.031) and first-test target-minus-other (−0.048) contrasts previously reported for this checkpoint. A visual routing block that is invariant to change status, invariant to tested location, and inversely related to stimulus presence is better described as a fixed contrast-dependent normalization than as spatial selection.

3.4 Memory keys: encode, maintain, reset, select

Everything that varies over time or across conditions is in the memory block (Figure 4). Four stages are visible.

Encode. At t_0 the memory map is exactly uniform (0.250 in all four quadrants, in all four conditions), as it must be while the recurrent state is still zero-initialized. By t_1 mass has moved onto *both* sample quadrants (0.43–0.49 each) leaving 0.04 in

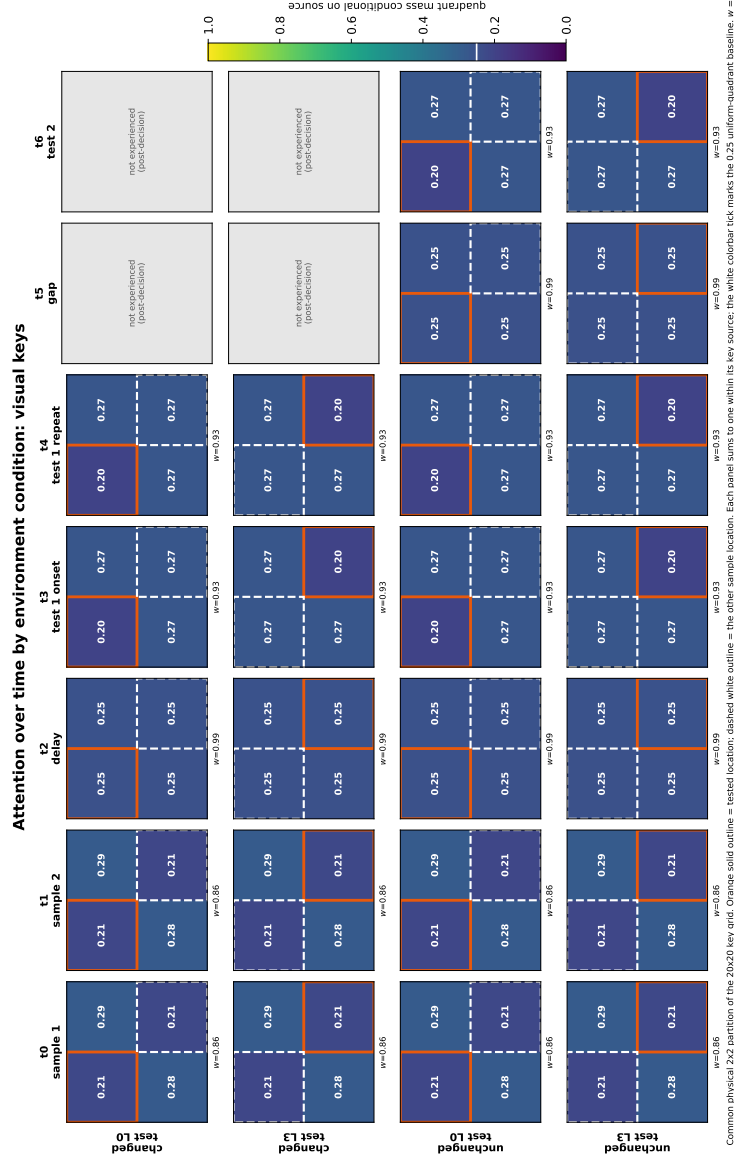


Figure 3: Visual keys carry no condition-contingent spatial signal. Rows are the four environment conditions; columns are the seven frames. Each panel is the common physical 2×2 partition of the 20×20 visual-key grid, coloured by quadrant mass conditional on the visual source (each panel sums to one; the white tick on the colourbar is the 0.25 uniform baseline). Orange solid outline marks the tested location, dashed white the other sample location, and w below each panel is the visual share of the joint softmax. Grey panels are frames that changed trials never experience, because the episode terminates in the first-test window. The map is identical across rows and is set entirely by which patches contain a Gabor — which it avoids.

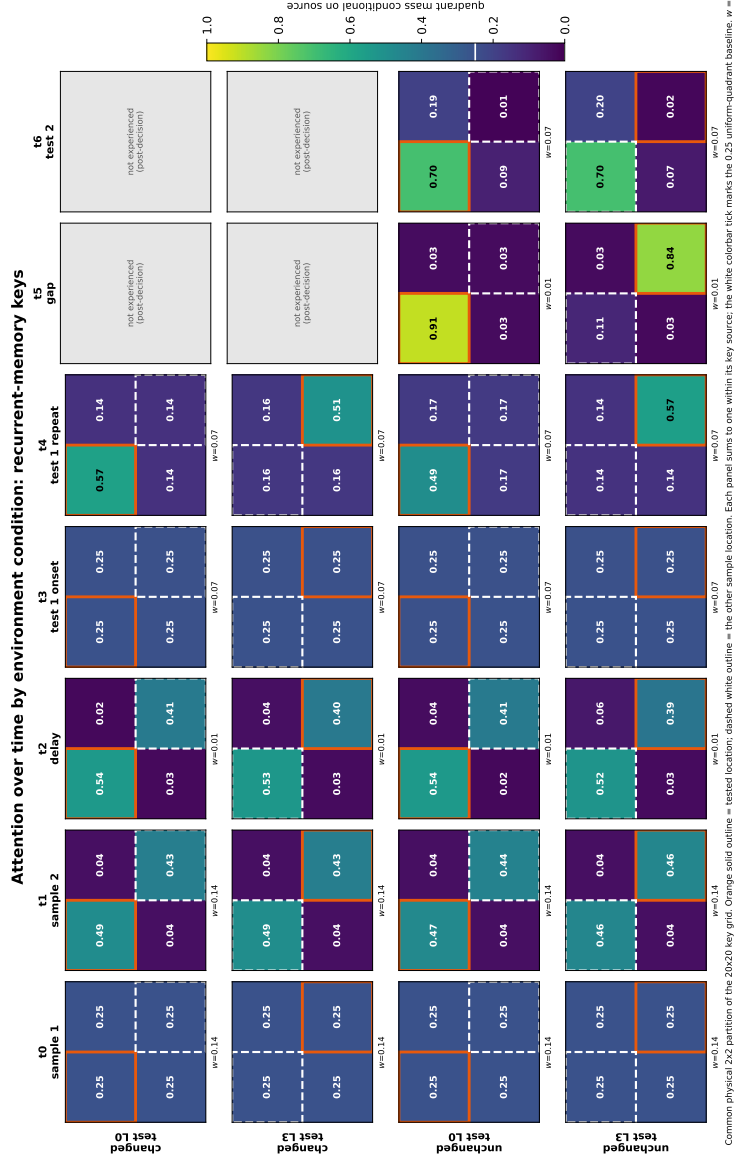


Figure 4: Recurrent-memory keys encode, maintain, reset, and select. Same layout, scores, and outlines as Figure 3, for the memory key block. Uniform at t_0 (the memory is zero-initialized), split across both sample quadrants at t_1 – t_2 , near-uniform again at first-test onset t_3 , tested-contingent at t_4 and through the gap t_5 , and — critically — locked to the top-left quadrant at t_6 irrespective of which location was actually tested, which is what produces the sign reversal in the tested-minus-other contrast.

Table 1: Tested-minus-other location contrasts by frame and condition, from the tested-location query. Visual routing is a constant -0.048 wherever a test stimulus is present; memory routing is tested-contingent at t_4 (and at t_5 , omitted here because only the unchanged conditions reach it) but location-locked at t_6 . Dashes are frames the condition never experiences.

Condition	Source	t_1	t_2	t_3	t_4	t_6
changed, test L0	visual	0.000	-0.000	-0.049	-0.048	—
	memory	+0.018	+0.001	0.000	+0.130	—
changed, test L3	visual	0.000	+0.000	-0.048	-0.048	—
	memory	-0.010	-0.001	-0.000	+0.094	—
unchanged, test L0	visual	-0.000	-0.000	-0.048	-0.048	-0.049
	memory	+0.012	-0.001	-0.000	+0.097	+0.194
unchanged, test L3	visual	0.000	+0.000	-0.048	-0.048	-0.048
	memory	+0.005	-0.002	-0.000	+0.116	-0.188

each blank quadrant. Both items are encoded, which is the correct behaviour for a task with no cue.

Maintain. At the blank delay t_2 the split persists (0.52–0.54 at L0, 0.39–0.41 at L3, 0.02–0.06 in the blanks). This map is *the same in all four conditions*, including the two whose tested location is L3. The asymmetry is a fixed location bias, not a prediction of the upcoming test — which is the correct control, since nothing in the display could support such a prediction.

Reset. At first-test onset t_3 the quadrant map returns to 0.250 in every quadrant and every condition, and the tested-query same-minus-other contrast is zero to floating-point precision in both sources. At token level the memory map at t_3 is not literally uniform (coefficient of variation 0.10 across the 400 tokens, against 7.6 at t_1), but it is near-uniform enough that all quadrant structure disappears. The maintained spatial code is discarded exactly when the test arrives, and rebuilt one frame later.

Select. At t_4 the memory concentrates on the tested location: 0.574 and 0.491 at L0, 0.510 and 0.566 at L3, against 0.14–0.17 at the other sample location. The tested-location query shows the same thing, sending +0.094 to +0.130 more memory mass to its own quadrant than to the other. On unchanged trials the concentration strengthens through the blank gap t_5 (0.907 at L0, 0.839 at L3) — the only frame at which routing is both strongly selective and almost entirely detached from the visual input, since w_M is only 0.010 there.

3.5 The selection is not location-general

The apparent “reselect the tested site” story fails its control at the second test. At t_6 both unchanged conditions route 0.70 of their memory mass to the *same* quadrant, L0 — which is the tested location in one condition and the untested location in the other. The tested-minus-other contrast is therefore +0.194 when L0 is tested and -0.188 when L3 is tested (Figure 5, bottom row, third and fourth columns; Table 1). Averaged over locations this cancels to nearly zero, which is exactly what the pooled estimate for this checkpoint reported (+0.0015, 95% CI [-0.050, +0.051], $n = 64$).

This matters for how the t_4 and t_5 results should be read. At those frames the selection genuinely follows the tested location and survives the L0/L3 control. At t_6 it does not, and a location-collapsed analysis would have hidden that by averaging a positive and a negative effect. **Reporting the tested-minus-other contrast separately at each tested location is not a robustness check here; it is the difference between a real result and an artefact.**

3.6 None of this is shown to be response-causal

A companion intervention on the same frozen checkpoint applied a -6 additive key-logit bias at the future-tested quadrant during the two sample frames. The manipulation worked as a routing manipulation: mean attention mass at that quadrant fell from 0.2342 to 0.0011. It did nothing behaviourally. Across 704 matched trial pairs there were *zero* discordant correctness outcomes, giving an observed Δ accuracy of +0.000; because the paired bootstrap is degenerate when every pair agrees, the conservative two-sided no-discordance bound is $|\Delta\text{accuracy}| < 0.0052$. The identical operation at the other active sample and at a blank control quadrant was equally inert. Hit rate, d' , and mean hit frame were unchanged at every probe magnitude.

Sample-phase spatial allocation in this checkpoint is therefore descriptive routing evidence and nothing more. The measured maps are real and highly reproducible under noise, but the one causal test we ran says the agent’s first-test decision does not depend on them.

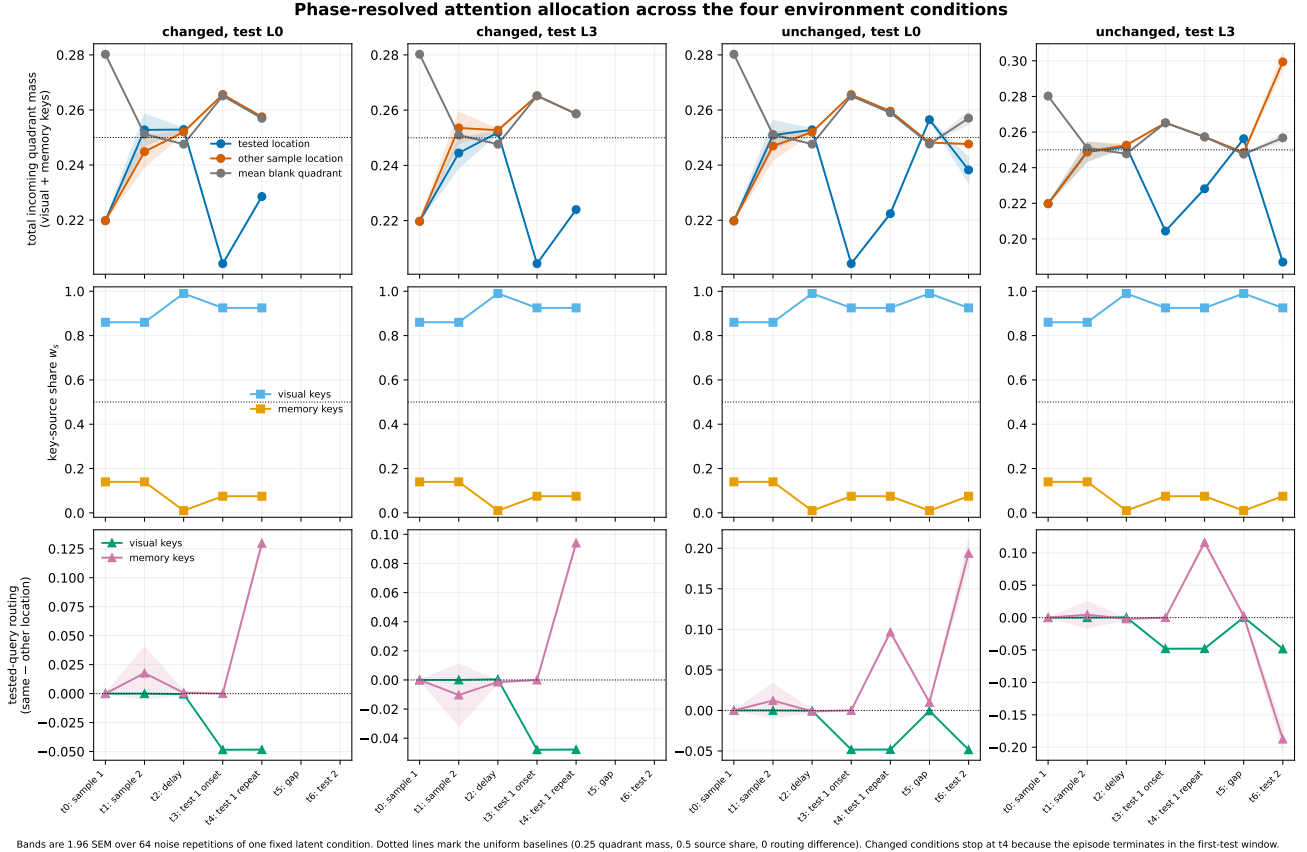


Figure 5: Phase-resolved allocation across the four environment conditions. Top: total incoming quadrant mass (visual + memory) at the tested location, the other sample location, and the mean blank quadrant; the dotted line is the 0.25 uniform baseline. The combined trace is dominated by the visual block and stays within ± 0.05 of uniform. Middle: key-source share, which tracks the number of visible Gabors. Bottom: tested-location query routing, same minus other location, per source; the memory trace rises at t_4 in all conditions but reverses sign between the two tested locations at t_6 . Bands are 1.96 SEM over 64 noise repetitions of one fixed latent condition.

4 Discussion

Three things are worth carrying forward.

Source separation is doing the work. If the visual and memory key blocks are summed before reduction — the co-located “V+M” diagnostic — the entire memory story is buried under a visual block that is 6 to 99 times larger and carries no condition-contingent signal. The combined trace in the top row of Figure 5 never leaves a ± 0.05 band around uniform, while the memory block underneath it swings from 0.25 to 0.91. The VDA-series convention of splitting sources before query reduction is what makes the effect visible, and this is a sharper demonstration of that than any cued case we have run.

The reset at t_3 is the most interesting unexplained observation. A recurrent memory that has held a two-item spatial code across a blank delay discards its quadrant structure at the exact frame the test appears, then rebuilds a one-item, tested-location code one frame later. We do not know whether this is a genuine computational stage — clear the maintained code, then write the comparison — or an incidental consequence of the memory update being dominated by the new visual input. Distinguishing those requires reading the recurrent state directly rather than the attention over it.

Uncued does not mean unselective, but it does raise the evidential bar. Without a cue there is no experimenter-named location to align routing to before the test, so the only defensible spatial contrasts are post-test ones, and they must be reported per tested location. The t_6 reversal shows what happens otherwise.

The obvious next experiments are (i) a corrected-contract retrain, without which none of the above can be promoted past provenance; (ii) clamping at t_4 – t_5 rather than at the sample frames, since that is where the tested-contingent signal actually is; and (iii) a linear probe on the recurrent state at t_2 and t_3 to test whether the maintained code really is discarded or merely becomes invisible to attention.

5 Validity and provenance

This checkpoint lineage is marked invalid for behavioural claims under the corrected Luo task contract. The enforced contract requires each sample orientation to be drawn independently from $\text{Uniform}[0^\circ, 180^\circ)$ on every trial, with the signed change drawn from $\text{Uniform}(-\theta, \theta)$ and θ controlling only the maximum absolute change. The frozen checkpoint records fixed session base orientations (98.79° at L0, 128.73° at L3) and carries no `orientation_sampling` marker, so it was trained under the earlier fixed-pair, curriculum-coupled generator. Its producer script now refuses that checkpoint outright. The consequences for this report are specific:

- Sample orientations were near-constant across training, so the recurrent memory had far less to encode than the corrected task demands. The encode/maintain stages in Figure 4 may reflect a memorized pair rather than trial-specific storage.
- No d' , criterion, psychometric, lesion, or Luo-replication claim from this lineage is admissible, including the behavioural curves in Figure 2, which are reproduced here only to locate the 18° attention probe.
- Warm-starting a corrected run from this checkpoint would contaminate it. A neutral parent must be retrained from scratch.

What does survive is the measurement methodology and the qualitative dissociation it exposes: source-split, per-frame, per-condition, survivor-controlled quadrant maps, displayed on a common physical partition with an explicit uniform baseline and an explicit source share. Those conventions transfer unchanged to the corrected retrain, and the analysis in this report can be regenerated against it by pointing the same producer at the new assay directory.

Additional boundaries that hold regardless of the contract issue: the intervals are over noise repetitions of one fixed latent condition at one checkpoint, not over training seeds or over the trial distribution; attention weights are normalized routing, not orientation-content decoding; the -6 logit bias is strong soft suppression rather than hard exclusion, leaving 0.48% of natural target-region mass; and the biological comparison target, Luo and Maunsell (2015) V4, is a reference point for the task design only — no neural-equivalence claim is made anywhere in this report.

Reproduction

All figures are produced by `analysis/make_luo2015_vda_style_attention_figs.py`, which reads the frozen `fixed_condition_attention.npz` cache from `reports/luo2015_scientific_assay_20260803_production` and writes `figure_statistics.json` alongside them. It re-runs no checkpoint and trains nothing. Every number quoted in this report appears in that JSON file or in the assay's `data/summary.json`.